2025.12.16 HW11 习题七

A3. (1) $L(\theta)=\prod_{i=1}^{n} f(x_i;\theta)=2^{-n\theta}\theta^{n}\left(\prod_{i=1}^{n} x_i\right)^{\theta-1}$

$\ln L(\theta)=-n\theta\ln 2+n\ln\theta+(\theta-1)\left(\sum_{i=1}^{n}\ln(x_i)\right)$

$\frac{d\ln L(\theta)}{d\theta}=-n\ln 2+\frac{n}{\theta}+\sum_{i=1}^{n}\ln(x_i)$

$\Rightarrow\hat{\theta}=\frac{1}{\ln 2-\frac{1}{n}\sum_{i=1}^{n}\ln x_i}$

代入得 0.577

(2) $L(\theta)=\frac{1}{(2\theta)^n}e^{-\sum_{i=1}^{n}\frac{|x_i|}{\theta}}$.

$\ln L(\theta)=-n\ln(2\theta)-\sum_{i=1}^{n}\frac{|x_i|}{\theta}$

$\frac{d\ln L(\theta)}{d\theta}=-\frac{n}{\theta}+\sum_{i=1}^{n}\frac{|x_i|}{\theta^2}=0\Rightarrow\hat{\theta}=\frac{1}{n}\sum_{i=1}^{n}|x_i|$.

代入得 0.468

(3) 显然 $\hat{\theta}=\min\{x_1,x_2\dots x_n\}$

代入得 $\hat{\theta}=0.35$

A7. $L(\theta) = \frac{1}{\theta^n}\prod_{i=1}^{n} x_i \cdot e^{-\sum_{i=1}^{n}\frac{x^2}{2\theta}}$.

则 $\ln L(\theta) = -n\ln\theta + \sum_{i=1}^{n}\ln x_i - \sum_{i=1}^{n}\frac{x_i^2}{2\theta}$.

$$\frac{d\ln L(\theta)}{d\theta} = -\frac{n}{\theta} + \sum_{i=1}^{n}\frac{x_i^2}{2\theta^2} = 0 \Rightarrow \hat{\theta} = \frac{A_2}{2}$$

代入 $\hat{\theta} = \frac{1}{2n}\sum_{i=1}^{n} x_i^2$.

$$\mu_2 = E(X^2) = \int_0^{\infty} x^2 \cdot \frac{x}{\theta} e^{-\frac{x^2}{2\theta}}\,dx = 2\theta.$$

$$\therefore \hat{\mu}_2 = 2\hat{\theta} = A_2 = \frac{1}{n}\sum_{i=1}^{n} x_i^2.$$

$$p = P(X>1) = \int_1^{\infty}\frac{x}{\theta}e^{-\frac{x^2}{2\theta}}\,dx = \exp\left(-\frac{1}{2\theta}\right).$$

$$\therefore \hat{p} = \exp\left(-\frac{1}{2\hat{\theta}}\right) = \exp\left(-\frac{1}{A_2}\right) = \exp\left(-\frac{n}{\sum_{i=1}^{n} x_i^2}\right)$$

A8 $E\left(a\sum_{i=1}^{9}(x_{i+1}-x_i)^2\right) = aE\left(\sum_{i=1}^{9}(x_{i+1}-\mu)^2 + \sum_{i=1}^{9}(x_i-\mu)^2 - 2\sum_{i=1}^{9}(x_{i+1}-\mu)(x_i-\mu)\right)$

$$= 18a\sigma^2 = \sigma^2 \Rightarrow a = \frac{1}{18}.$$

B2. 记上述事件为A，则

$$P(A;N)=\frac{C_r^t C_{N-r}^{S-t}}{C_N^S}$$

$$\frac{P(A;N+1)}{P(A;N)}=\frac{(N+1-r)(N+1-S)}{(N+1)(N+1-r-S+t)}\geq 1.$$

$$\Rightarrow rS\geq t(N+1) \quad \therefore N+1\leq \frac{rS}{t}$$

故 $\hat{N}=\left[\frac{rS}{t}\right]$.

B3. $E(X)=a_1\theta+a_2\cdot\frac{1-\theta}{2}+a_3\cdot\frac{1-\theta}{2}=\mu_1$.

$$\Rightarrow \hat{\theta}=\frac{2\bar{X}-(a_2+a_3)}{2a_1-a_2-a_3}$$

$$L(\theta)=\theta^{n_1}\left(\frac{1-\theta}{2}\right)^{n_2}\left(\frac{1-\theta}{2}\right)^{n_3}$$

$$\ln L(\theta)=n_1\ln\theta+(n_2+n_3)\cdot\ln\frac{1-\theta}{2}$$

$$\frac{d\ln L(\theta)}{d\theta}=\frac{n_1}{\theta}-\frac{n_2+n_3}{1-\theta}=0\Rightarrow\hat{\theta}=\frac{n_1}{n}.$$

B6. (1) $E(X)=\int_0^{\infty}\frac{x}{\theta}e^{-\frac{x}{\theta}}dx=\theta.$

$\therefore \hat{\theta}=\overline{X}.$

$L(\theta)=\prod_{i=1}^{n}f(x;\theta)=\frac{1}{\theta^n}\cdot e^{-\sum_{i=1}^{n}\frac{x_i}{\theta}}.$

$\ln L(\theta)=-n\ln\theta-\sum_{i=1}^{n}\frac{x_i}{\theta}.$

$\frac{d\ln L(\theta)}{d\theta}=-\frac{n}{\theta}+\sum_{i=1}^{n}\frac{x_i}{\theta^2}=0 \Rightarrow \hat{\theta}=\frac{1}{n}\sum_{i=1}^{n}x_i=\overline{X}.$

(2) $\hat{\theta}_c=c.\sum_{i=1}^{n}X_i.$ 记 $T=\sum_{i=1}^{n}X_i.$

$$\begin{aligned}MSE(c)=E((cT-\theta)^2)&=Var(cT)+(E(cT-\theta))^2\\&=c^2n\theta^2+(cn\theta-\theta)^2\\&=\theta^2(c^2n+(cn-1)^2)\end{aligned}$$

令 $g(c)=c^2n+(cn-1)^2$, 则 $g'(c)=2c(n+n^2)-2n=0$

$\Rightarrow c=\frac{1}{n+1}$

$\therefore$ c 在 $\frac{1}{n+1}$ 时，均方误差规则下最优.

(3) $\hat{\theta}_c=\frac{1}{n+1}\sum_{i=1}^{n}X_i$ $\quad E(\hat{\theta}_c)=\frac{1}{n+1}n\theta\xrightarrow{P}\theta$, $n\to\infty$.

$\therefore$ 是 θ 的相合估计量. $\quad Var(\hat{\theta}_c)=\left(\frac{1}{n+1}\right)^2n\theta^2\xrightarrow{P}0.$, $n\to\infty$.

B7. (1) $f(x;\theta)=\begin{cases}\dfrac{3x^2}{\theta^3}, & 0<x<\theta.\\ 0, & 其他.\end{cases}$

$E(X)=\int_0^\theta \frac{3x^3}{\theta^3}dx=\frac{3\theta}{4} \Rightarrow \hat{\theta}=\frac{4}{3}\overline{X}.$

$E(\hat{\theta})=E(\frac{4}{3}\overline{X})=\frac{4}{3}E(\overline{X})=\frac{4}{3}E(X)=\theta.$

$\therefore \hat{\theta}$ 是 θ 的无偏估计

(2) $f(x;\lambda)=\begin{cases}\dfrac{\lambda x^{\lambda-1}}{3^\lambda}, & 0<x<3.\\ 0, & 其他\end{cases}$

$L(\lambda)=\prod_{i=1}^{n}\left(\frac{\lambda x_i^{\lambda-1}}{3^\lambda}\right).$

$\ln L(\lambda)=n\ln\lambda+(\lambda-1)\sum_{i=1}^{n}\ln x_i-n\lambda\ln 3.$

$\frac{d\ln L(\lambda)}{d\lambda}=\frac{n}{\lambda}+\sum_{i=1}^{n}\ln x_i-n\ln 3=0 \Rightarrow$

$\hat{\lambda}=\dfrac{1}{\ln 3-\frac{1}{n}\sum_{i=1}^{n}\ln x_i}$，它是 λ 的相合估计，因为

$\frac{1}{n}\sum_{i=1}^{n}\ln x_i \xrightarrow{P} E(\ln X)=\ln 3-\frac{1}{\lambda}$，$\hat{\lambda}\xrightarrow{P}\lambda$，$n\to\infty$.